Unit 6, Lesson 5: Connecting Ratios and Rates

Benchmark

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

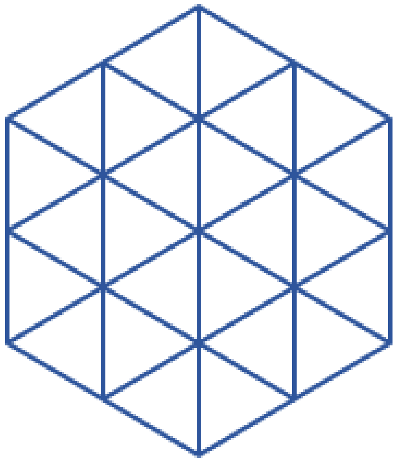
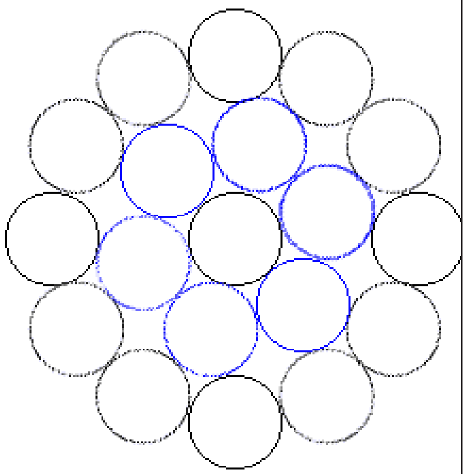
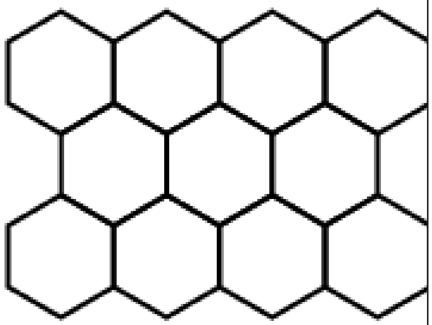
Learning Target

☐ I can make connections between ratios, rates, and unit rates.

6.5.1 Warm-Up: Why Do Honeybees Love Hexagons?

Honeybees are actually amazing mathematicians. Scientists claim they can even calculate angles. They also live inside one of the most mathematically efficient architectural designs there is: the beehive. It takes a lot of work for bees to produce wax. The hexagonal shape of the cells in a honeycomb allows bees to store the most amount of honey while using the least amount of wax. This design likely evolved over time.

Consider the shapes used in the designs shown.

Triangular Cells	Circular Cells	Hexagonal Cells
		

1. How do the designs demonstrate that the hexagonal shape allows bees to store the most amount of honey using the least amount of wax?

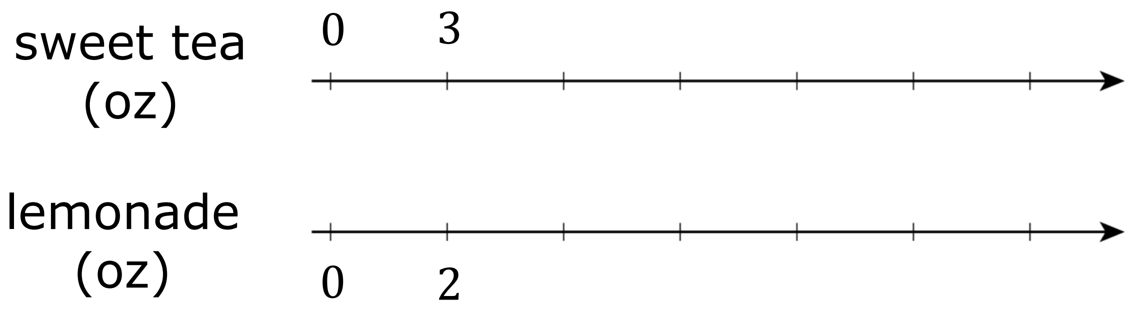
2. This is an example of mathematics in nature. What is another example of mathematics in nature you can think of?

6.5.2 Guided Instruction: Connecting Ratios and Rates

A given ratio or rate can be used to generate many equivalent ratios, which can be helpful in solving problems. Double number lines, tape diagrams, mixture charts, and ratio tables are useful tools for finding equivalent ratios.

1. Noah's favorite drink is a mixture of sweet tea and lemonade. He uses the ratio of 3 ounces (oz) of sweet tea to 2 oz of lemonade to make it.

   A double number line diagram is shown to represent the situation. Complete the double number line diagram.



A table can also be used to represent this relationship.

2. Complete the table to show how many ounces of each beverage are needed to make different-sized batches of Noah's favorite drink.

Sweet Tea (oz)	**Lemonade (oz)**
3	2
21	
	10
4.5	
	32
1.5	

Recipes are scaled up to make larger batches with the same taste. They can also be scaled down to make smaller batches with the same taste. Equivalent ratios can be used to determine the amount of ingredients needed when scaling a recipe up or down.

3. When scaling a recipe up, such as to find how many ounces of sweet tea to mix with 32 ounces of lemonade, which operation is used?

4. When scaling a recipe down, such as to find how many ounces of lemonade to mix with 1.5 ounces of sweet tea, which operation is used?

5. William noticed he could use similar thinking with the double number line diagram. He drew an arrow from 3 to 15 on the top number line and another arrow from 2 to 10 on the bottom number line. Complete the statement to describe what he saw.

   William noticed the ratio $\frac{3}{2}$ can be used to find the ratio $\frac{15}{10}$ by

   - ○ adding 12 to both parts of the ratio.
   - ○ multiplying both parts of the ratio by 5.
   - ○ dividing both parts of the ratio by 5.

Equivalent ratios can be found by multiplying or dividing each part of a ratio by the same value.

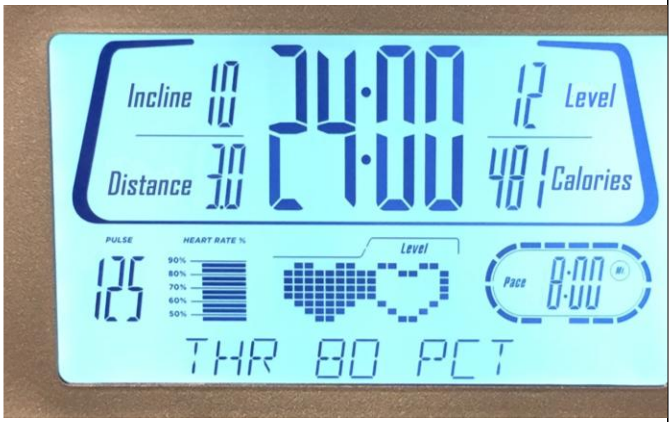
6. Show how the ratios $3{:}2$ and $48{:}32$ from the table are equivalent, using either multiplication or division.

7. Shay noticed the table of Noah's favorite drink included a unit rate. Circle the unit rate in the table. Then, complete the statement to explain its meaning in context.

   This unit rate means there are _____ ounces of ____________________ in the drink for each ounce of ____________________.

6.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed

Mai and Jada each ran on a treadmill. The treadmill display shows the distance, in miles, each person ran and the amount of time it took them, in minutes and seconds.

<table>
<tr><th>Mai’s Treadmill Display</th><th>Jada’s Treadmill Display</th></tr>
<tr><td>


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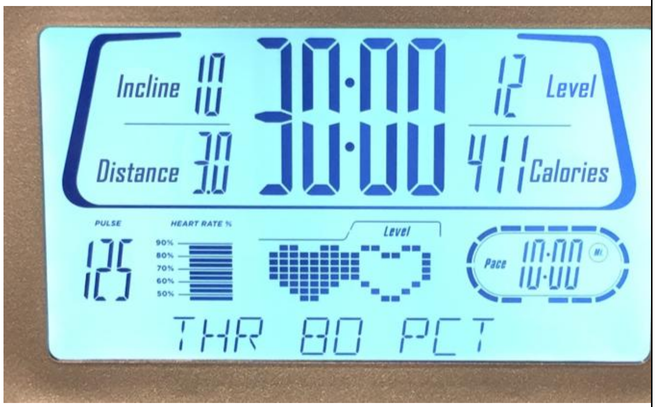

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</table>

1. Discuss with your partner the similarities and differences between Mai’s and Jada’s workouts. Write them in the table below.

Similarities	Differences

2. Express each person’s workout as a ratio of distance to time, including units.

	Ratio of Distance to Time
Mai’s Workout	
Jada’s Workout	

3. Each person ran at a constant rate by setting the speed on their treadmill. Discuss with your partner who ran at a faster rate using the ratios.

4. Eli thought it would be helpful to rewrite the ratios as unit rates when deciding who ran faster. His incomplete work is shown. Work with your partner to complete his work.

Mai's Workout	**Jada's Workout**
24 min : 3 mi ÷3 ÷3 8 min : 1 mi ÷8 ÷8 _____ : _____	30 min : 3 mi ÷3 ÷3 10 min : 1 mi ÷10 ÷10 _____ : _____

5. Complete the statements.

It took Mai _____ minutes to run 1 mile, and it took Jada _____ minutes to run 1 mile. This shows that
○ Mai
○ Jada
ran faster because it took her
○ less
○ more
time to run the same distance.

In 1 minute, Mai ran _____ mile(s) and Jada ran _____ mile(s). This shows that
○ Mai
○ Jada
ran faster because she ran
○ less
○ more
distance in the same amount of time.

6.5.4 Try It: Rates and Ratios in Context of Speed

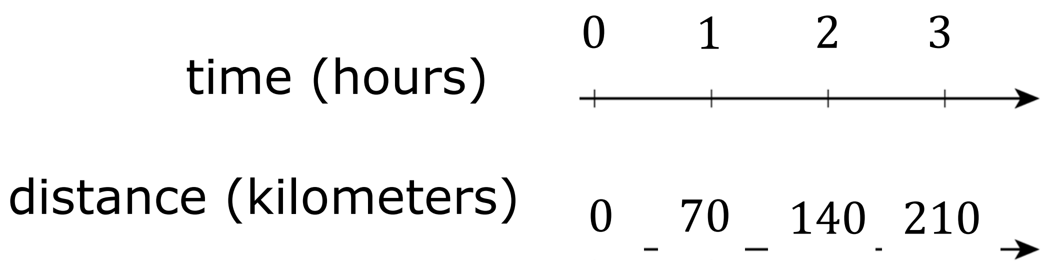
1. A slug travels 3 centimeters in 3 seconds. A snail travels 6 centimeters in 6 seconds. Ethan says, “The snail was traveling faster because it went a greater distance.”

   a. Discuss Ethan’s statement with your partner.

   b. Using ratios or rates, explain whether you agree with Ethan.

6.5.5 Your Turn!

1. Nadia is driving at a constant speed on a long stretch of highway on a long drive from Melbourne, Florida, to Atlanta, Georgia. Her distance traveled over time is shown on the double number line.



How far did Nadia travel at this speed for 5 hours? Explain or show your reasoning.

2. Two cleaning solutions were created to clean tile floors as shown.

Mixture A	Mixture B
5 cups of water 2 cups of vinegar	15 cups of water 8 cups of vinegar

a. Explain why the mixtures will not result in the same cleaning solution.

b. Explain which solution will have a stronger smell of vinegar.

6.5.6 Wrap-Up: Growth Mindset

Complete each statement.

1. Today I am still unsure about …

2. To fill this gap, I intend to …